

Chapter 5g: Fixed Point Theorems in Banach Algebra and Lattice-Theoretic Structures

P-Lipschitzian Maps, Lattice Fixed Points, and Cascading Nonexpansive Maps

From Pathak's *An Introduction to Nonlinear Analysis and Fixed Point Theory*

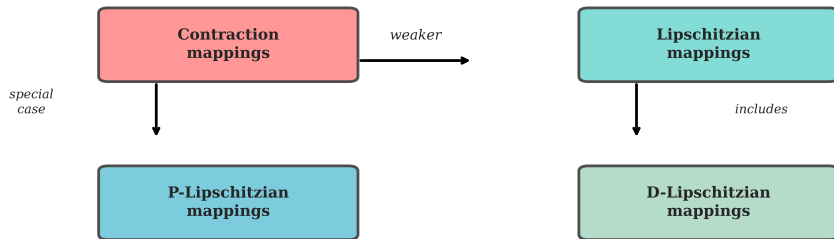
Outline

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- **Section 5.8:** Fixed Point Theorems in Banach Algebra
 - P-Lipschitzian Maps (Definition 5.94)
 - Fixed point existence for multiple operators
- **Section 5.9:** Lattice-Theoretic Fixed Point Theorems
 - Tarski's fixed point theorem
 - Increasing functions on lattices
- **Section 5.9.1:** Reflexivity and Perturbed Fixed Point Property
 - James's distortion theorems
 - Cascading nonexpansive maps

Mapping Classification

Hierarchy of Mapping Types



Definition 5.94: P-Lipschitzian Maps

Definition

A mapping $T : X \rightarrow X$ on a Banach space X is called **P-Lipschitzian** if there exists a nondecreasing function $\phi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$\|Tx - Ty\| \leq \phi(\|x - y\|), \quad \forall x, y \in X$$

Key Observations:

- Every D-Lipschitzian mapping is P-Lipschitzian
- The converse is *not* true in general
- More general than classical contractions

Theorem 5.183: Operator Equation with P-Lipschitzian Operators

Theorem

Let S be a closed, convex and bounded subset of a Banach algebra X , and let $A, C : X \rightarrow X$ and $B : S \rightarrow X$ be three operators such that

- (a) A and C are P -Lipschitzian with P functions ϕ_A and ϕ_C ;
- (b) B is completely continuous; and
- (c) $x = AxBy + Cx \Rightarrow x \in S, \forall y \in S$.

Then $AxBx + Cx = x$ has a solution whenever $M\phi_A(r) + \phi_C(r) < r, r > 0$, where $M = \|B(S)\|$.

Theorems 5.184 and 5.185: D-Lipschitzian Variants

Theorem (Theorem 5.184)

Let S be a closed, convex and bounded subset of a Banach algebra X , and let $A : X \rightarrow X$ and $B : S \rightarrow X$ be two operators such that

- (a) A is D -Lipschitzian;
- (b) $(I/A)^{-1}$ exists on $B(S)$; and
- (c) $x = AxBy \Rightarrow x \in S, \forall y \in S$.

Then $AxBx = x$ has a solution whenever $M\phi(r) < r$, where $M = \|B(S)\|$.

Proposition 5.14: Sufficient Condition

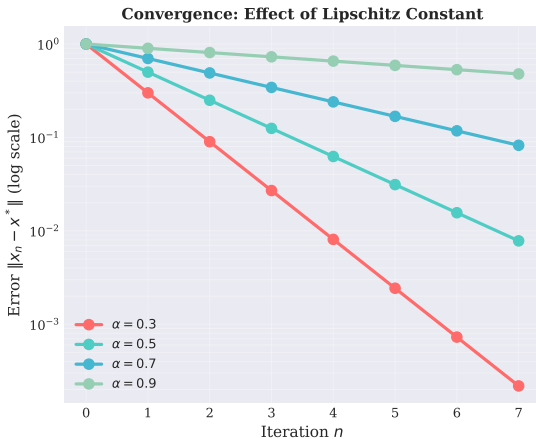
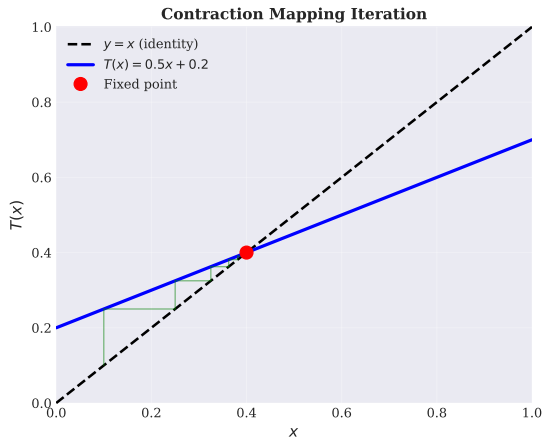
Theorem

Let $S = \{y \in X : \|y\| \leq r\}$ be a closed ball in Banach algebra X . Let $A, C : X \rightarrow X$ and $B : S \rightarrow S$ be operators satisfying hypothesis (a)–(b) of Theorem 5.177. If

$$\left\| \left(\frac{I - C}{A} \right) x \right\| \leq \|x\|$$

for all $x \in X$, then $x \in S$.

Visualization: Fixed Point Iteration



Banach Lattice Structure

Banach Lattice Structure

▼	Vector Space Structure
$\ \cdot\ $	Norm Structure
$\leq$	Lattice Order Structure
$\Leftrightarrow$	Norm-Lattice Compatibility

Key Definitions

- Fixed point: $f(x^*) = x^*$
- Increasing function: $x \leq y \implies f(x) \leq f(y)$
- Lattice completeness: every subset has supremum and infimum

Theorem 5.186: Tarski's Fixed Point Theorem

Theorem (Tarski)

Let $\mathfrak{A} = (A, \leq)$ be a complete lattice and f be an increasing function on A to A . Let P be the set of all fixed points of f .

Then P is not empty, and $(P, \leq)$ is a complete lattice; in particular,

$$\bigcup P = \bigcup \{x \in A : f(x) \geq x\} \in P$$

and

$$\bigcap P = \bigcap \{x \in A : f(x) \leq x\} \in P.$$

Key Ideas:

- No contractivity assumption!
- Applies to arbitrary increasing functions
- Fixed points form a complete lattice

Theorem 5.187: Generalized Lattice-Theoretic Result

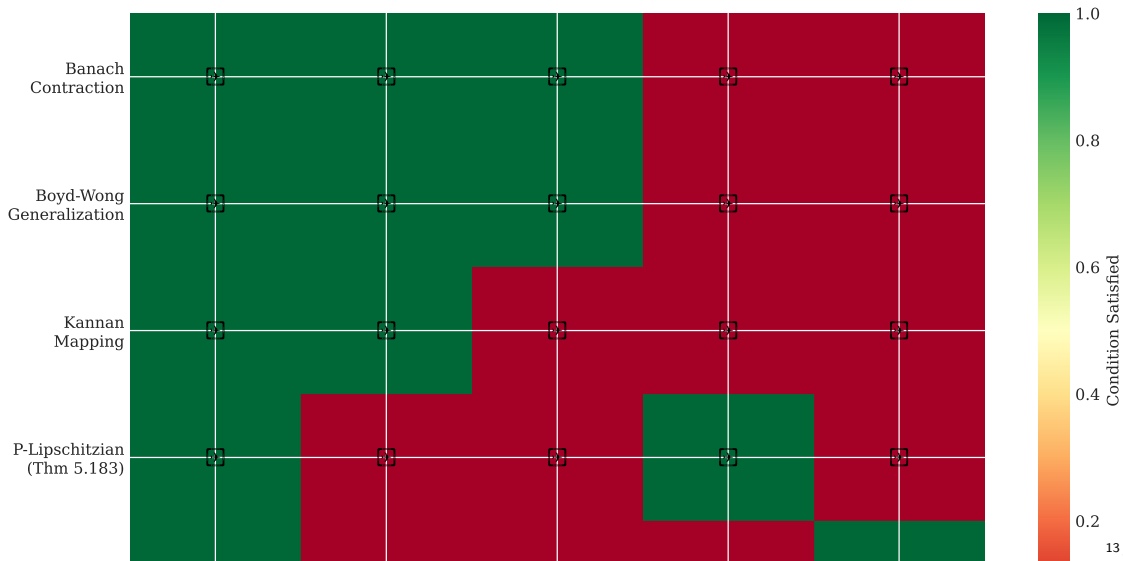
Theorem

Let $\mathfrak{A} = (A, \leq)$ be a complete lattice and F be a commutative set of increasing functions on A to A . Let P be the set of all common fixed points of all functions $f \in F$. Then P is not empty and $(P, \leq)$ is a complete lattice.

Extension: Allows multiple *commuting* functions; their common fixed points form a complete lattice.

Theorem Comparison

Comparison of Fixed Point Theorem Conditions



Cascading Nonexpansive Mappings

Definition

A mapping $T : C \rightarrow C$ on bounded, closed and convex subset C is **cascading nonexpansive** if there exists $(k_n) \in [1, \infty)$ with $k_n \rightarrow 1$ such that for all $n \in \mathbb{N}$ and $x, y \in C_n$,

$$\|Tx - Ty\| \leq k_n \|x - y\|$$

where $C_0 := C$ and $C_n := \overline{\text{co}}(T(C_{n-1})) \subseteq C_{n-1}$.

Key Properties:

- Norm-to-norm continuous
- Generalizes nonexpansive maps
- Arises naturally in ℓ_1 and c_0 spaces

James's Distortion Theorems

Theorem (James's Distortion Theorem for ℓ_1)

A Banach space X contains an isomorphic copy of ℓ_1 iff for every null sequence (ε_n) in $(0, 1)$, there exists $(x_n) \subseteq X$ such that

$$(1 - \varepsilon_n) \sum_{n=k}^{\infty} |t_n| \leq \left\| \sum_{n=k}^{\infty} t_n x_n \right\| \leq \sum_{n=k}^{\infty} |t_n|$$

for all $(t_n) \in \ell_1$ and $k \in \mathbb{N}$.

Theorem 5.188: Fixed Point Freeness

Theorem (Theorem 5.188)

*Let $(X, \|\cdot\|)$ be a Banach space containing an isomorphic copy of ℓ_1 or c_0 . Then there exists a bounded, closed and convex set $C \subseteq X$ and an affine cascading nonexpansive mapping $T : C \rightarrow C$ that is **fixed point free**.*

Implication: Non-reflexive spaces may fail fixed point property!

Theorems 5.189 and 5.190: Reflexivity Characterization

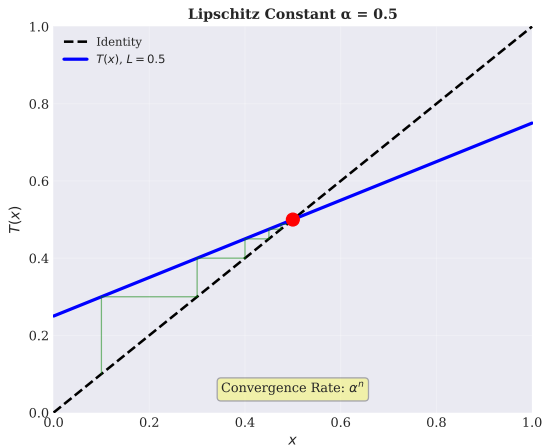
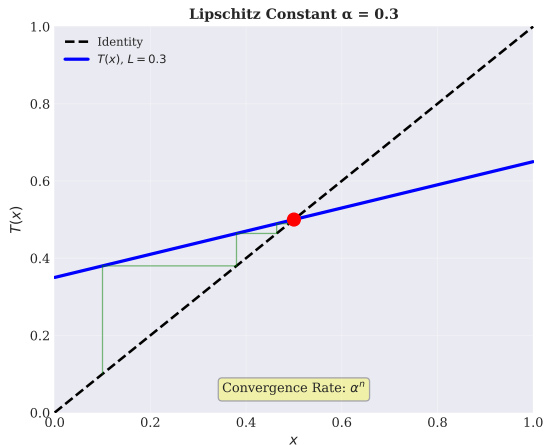
Theorem (Theorem 5.189)

Let $(X, \|\cdot\|)$ be a reflexive Banach space. Then there exists an equivalent norm $\|\cdot\|_-$ on X such that for every bounded, closed and convex subset $C \subseteq X$, every $\|\cdot\|_-$ -cascading nonexpansive mapping $T : C \rightarrow C$ has a fixed point in C .

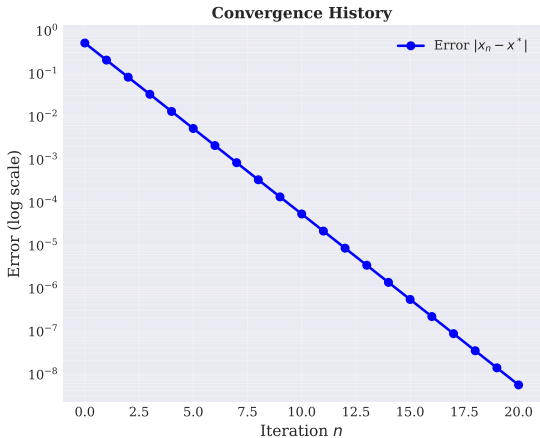
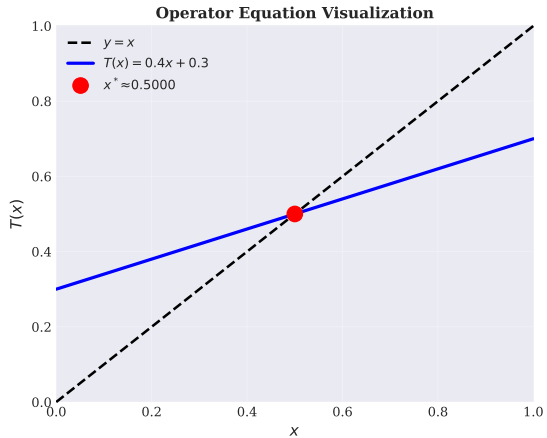
Theorem (Theorem 5.190)

For a Banach lattice (or space with unconditional basis, or operator ideal), reflexivity is equivalent to the fixed point property for cascading nonexpansive maps.

Lipschitz Constant Effect on Convergence



Numerical Example



Key Applications

- **Operator Equations:** Solving $AxBx + Cx = x$ in nonlinear systems
- **Lattice Optimization:** Finding extremal fixed points in order-theoretic structures
- **Banach Space Geometry:** Understanding spaces via fixed point properties
- **Differential and Integral Equations:** Existence and uniqueness of solutions with special structure
- **Iterative Methods:** Design and convergence analysis of numerical schemes

Important Distinctions

- **Contraction Mappings**

- Unique fixed point, exponential convergence

- **P-Lipschitzian Maps**

- General function condition, applies to multiple operators

- **Nonexpansive Maps**

- Requires geometric conditions, more general setting

- **Lattice-Theoretic**

- Guaranteed existence, no metric required

Key Theorems Summary

- **Theorem 5.183–5.185:** P-Lipschitzian and D-Lipschitzian operators in Banach algebras with existence conditions
- **Theorem 5.186–5.187:** Tarski's lattice-theoretic theorems with no contractivity assumptions
- **Theorems 5.188–5.190:** Reflexivity characterization via cascading nonexpansive maps in Banach lattices
- **James's Distortion Theorems:** Characterization of ℓ_1 and c_0 in Banach spaces

Questions?

Chapter 5g: Fixed Point Theorems in Banach Algebra and Lattice Structures